



Mock for End Term



The due date for submitting this assignment has passed.

Due on 2026-05-08, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

2 points

Choose all the subspaces isomorphic to R^4 .

☐

$$W_1 = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} : a + b + c + d + e + f = 0 \right\} \quad W_1 = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} : a + b + c + d + e + f = 0 \right\}.$$

☐

$$W_2 = \{(x, y, z, w, u) : x + y + z = w, u = 0\} \quad W_2 = \{(x, y, z, w, u) : x + y + z = w, u = 0\}.$$

☐

$$W_3 = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix} : a + b + c + d = 0, e + f + g + h = 0 \right\} \quad W_3 = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix} : a + b + c + d = 0, e + f + g + h = 0 \right\}.$$

☐

$$W_4 \text{ is the column space of the matrix } \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

☐

W_5 is the orthogonal complement of the zero subspace in R^4 with respect to the standard inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$W_4 \text{ is the column space of the matrix } \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

W_5 is the orthogonal complement of the zero subspace in R^4 with respect to the standard inner product.

2 points

Consider a basis $\{(1, 0, 1), (-1, 1, 1)\}$ of a subspace W in R^3 with the standard dot product. Find the projection vector of $(1, 2, 3)$ onto W .

☐

$$\left(\frac{2}{3}, \frac{4}{3}, \frac{10}{3}\right)$$

☐

$$\left(-\frac{2}{3}, -\frac{4}{3}, -\frac{10}{3}\right)$$

☐


$$\left(-\frac{2}{3}, \frac{4}{3}, -\frac{10}{3}\right)(-32, 34, -310)$$

☐

$$\left(\frac{2}{3}, -\frac{4}{3}, \frac{10}{3}\right)(32, -34, 310)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\left(\frac{2}{3}, \frac{4}{3}, \frac{10}{3}\right)(32, 34, 310)$$

Consider the vector space R^3 with usual addition and scalar multiplication. Let S be its subset defined as follows:

$$S = \{(1, 0, 1), (0, 1, 1)\}$$

and W_1 and W_2 be its vector subspaces defined as:

$$W_1 = \{(x, y, z) \mid x = z, \text{ and } y = z; x, y, z \in R\}$$

$$W_2 = \{(x, y, z) \mid z = x + y; x, y, z \in R\}$$

Answer the following questions:

2 points

Which of the following options is (are) true?

☐

S is linearly independent.

☐

S spans the vector space R^3 (with usual addition and scalar multiplication).

☐

$\text{Span}(S)$ is a proper subspace of R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent.

$\text{Span}(S)$ is a proper subspace of R^3 .

1 point

Which of the following options is (are) correct?

☐

S is a basis of W_1 .

☐

S is a basis of W_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a basis of W_2 .

What is the dimension of W_1 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

What is the dimension of W_2 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Consider a map $T: V \rightarrow W$, defined by $T(A) = \text{trace}(A)$, where V is the vector space of all matrices of order 2 with respect to the usual addition and scalar multiplication in matrices, and W is $\mathbb{R}$ with usual addition and scalar multiplication. Let $\beta = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ be an ordered basis of V and $\gamma = \{1\}$ be an ordered basis of W . Answer the following questions using the information above.

2 points

Choose the set of correct options from the following:

☐

$T(A + B) = T(A) + T(B)$ for all $A, B \in V$.

☐

$T(cA) = cT(A)$ for all $A \in V$, and $c \in \mathbb{R}$.

☐

T is not a linear transformation.

☐

$T(I) = 0$, where I denotes the identity matrix of order 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(A + B) = T(A) + T(B)$ for all $A, B \in V$.

$T(cA) = cT(A)$ for all $A \in V$, and $c \in \mathbb{R}$.

2 points

Choose the set of correct options from the following:

☐

If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

☐

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☐

If T is a linear transformation and M is the matrix representation of T with respect to the bases β for the domain and γ for the codomain, then $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If T is a linear transformation and M is the matrix representation of T with respect to the bases β

β for the domain and $\gamma\gamma$ for the codomain, then $M = [1001]M = [1001]$.

3 points

Let AA and BB be 4×4 real matrices such that $\text{rank}(A) = 2$ and $\text{rank}(B) = 3$. Choose all the correct options from the following:

☐

AA and BB are not equivalent.

☐

The system $Ax = 0$ has infinitely many solutions and the solution space has dimension 2.

☐

The nullity of B is 1, and hence $Bx = 0$ has a unique non-trivial solution up to scalar multiples.

☐

The columns of A lie in a subspace of $\mathbb{R}^4$ of dimension 2, while the columns of B span $\mathbb{R}^4$.

☐

If u_1, u_2, u_3, u_4 are the columns of B , then these vectors must be linearly dependent but cannot all lie in the same plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA and BB are not equivalent.

The system $Ax = 0$ has infinitely many solutions and the solution space has dimension 2.

The nullity of B is 1, and hence $Bx = 0$ has a unique non-trivial solution up to scalar multiples.

If u_1, u_2, u_3, u_4 are the columns of B , then these vectors must be linearly dependent but cannot all lie in the same plane.

3 points

Choose the correct statements for the function $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

$f(x, y) = 0$ for $(x, y) = (0, 0)$

☐

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$

☐

f is not continuous at $(0, 0)$.

☐

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.

☐

$\lim_{(x, y) \rightarrow (1, -1)} f(x, y) = \frac{1}{2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is not continuous at $(0, 0)$.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.

$\lim_{(x, y) \rightarrow (1, -1)} f(x, y) = \frac{1}{2}$

2 points

From the options below, choose all the options containing functions defined on the main $\mathbb{R}^2 \setminus \{(0,0)\}$ for which the limit of the function at the origin along any straight through the origin exists, and the limits are all equal.

☐ $f(x, y) = \frac{(x-y)^2}{x^2+y^2}$

☐ $f(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$

☐ $f(x, y) = (x^2 + y^2) \sin\left(\frac{1}{x^2+y^2}\right)$

☐ $f(x, y) = \frac{xy^2}{x^2+y^4}$

☐ $f(x, y) = \frac{x^2-y^2}{x^2+y^2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = \frac{xy}{\sqrt{x^2+y^2}}$$

$$f(x, y) = (x^2 + y^2) \sin\left(\frac{1}{x^2+y^2}\right)$$

$$f(x, y) = \frac{xy^2}{x^2+y^4}$$

2 points

Consider the following functions f defined as:

$$f(x, y, z) = xyz. f(x, y, z) = xyz.$$

Then the tangent line at the point $(1,0,1)$ in the direction of $(1,1,1)$ is parallel to the line



$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}} x(t) = 1 + \frac{t}{3}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3} x(t) = 1 + \frac{t}{3}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



$$x(t) = 1 + \frac{t}{3}, y(t) = \frac{t}{3}, z(t) = 1 + \frac{t}{3}, u(t) = \frac{t}{3} x(t) = 1 + \frac{t}{9}$$



☐

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$y(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$z(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$u(t) = \frac{t}{\sqrt{3}}$$

☐
☐

None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}x(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$y(t) = \frac{t}{\sqrt{3}}$$

☐

$$z(t) = 1 + \frac{t}{\sqrt{3}}$$

☐

$$u(t) = \frac{t}{\sqrt{3}}$$

☐
☐

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be the real-valued function defined by

$$f(x, y) = x^2 - xy + y^2, \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Let $T(3, 1)$ denote the tangent plane to the graph of f at the point $(3, 1)$. Since $T(3, 1)$ is an affine subspace of $\mathbb{R}^3$, there exists a vector $\mathbf{v} \in \mathbb{R}^3$ and a unique subspace $U \subseteq \mathbb{R}^3$ such that $T(3, 1) = \mathbf{v} + U = \{\mathbf{v} + \mathbf{u} \mid \mathbf{u} \in U\}$.

2 points

Which of the following are possible candidates for $\mathbf{v}$?

☐

$$(2, 1, 2)$$

☐

$$(0, 7, 0)$$

☐

$$(1, -2, 0)$$

☐

$$(2, 0, -1)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(2, 1, 2)$$

$$(1, -2, 0)$$

2 points

Which of the following is a basis for U ?

☐ $\{(1, -1, -1), (1, 3, 2)\} \{(1, -1, -1), (1, 3, 2)\}$ ☐ $\{(1, 2, 3), (-1, -1, -1)\} \{(1, 2, 3), (-1, -1, -1)\}$ ☐ $\{(0, 1, -1), (0, -1, 1)\} \{(0, 1, -1), (0, -1, 1)\}$ ☐ $\{(1, 2, 3), (0, -1, 1)\} \{(1, 2, 3), (0, -1, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

 $\{(1, 2, 3), (0, -1, 1)\} \{(1, 2, 3), (0, -1, 1)\}$

Consider the function $f(x, y) = x^3 - 3x + y^3 - 3y^2$ $f(x, y) = x^3 - 3x + y^3 - 3y^2$.

2 points

Which of the following points are the critical points of f ?

☐ $(1, 0)$ ☐ $(0, 0)$ ☐ $(-1, 0)$ ☐ $(0, 1)$ ☐ $(1, 2)$ ☐ $(2, 1)$ ☐ $(-1, 2)$ ☐ $(2, -1)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

 $(1, 0)$ $(-1, 0)$ $(1, 2)$ $(-1, 2)$

2 points

Choose the correct options from the following.

☐ $(1, 0)$ is a local minima of the function f .☐ $(0, 0)$ is a local minima of the function f .☐ $(1, 2)$ is a local minima of the function f .☐ $(-1, 2)$ is a saddle point of the function f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

 $(1, 2)$ is a local minima of the function f . $(-1, 2)$ is a saddle point of the function f .

2 points

Choose the correct options from the following.

☐ $(-1, 0)$ is a local maxima of the function f .

☐

(0,1) is a local maxima of the function f .

☐

(1,0) is a saddle point of the function f .

☐

(2,1) is a local maxima of the function f .

☐

(2,-1) is a local maxima of the function f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(-1,0) is a local maxima of the function f .

(1,0) is a saddle point of the function f .

Let A_1 and A_2 be two affine spaces defined by the solution space of the system of linear equations

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

respectively. Let V_1 and V_2 be the vector subspaces of R^3 corresponding to A_1 and A_2 . Answer the following questions using the information given above.

3 points

Choose the correct options.

☐

V_1 has dimension 1, where as the dimension of V_2 is 2.

☐

V_1 has dimension 2, where as the dimension of V_2 is 1.

☐

Both the V_1 and V_2 have dimension 3.

☐

Both the V_1 and V_2 have dimension 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_1 has dimension 1, where as the dimension of V_2 is 2.

3 points

Which of the following function can represent a map from A_2 to A_1 correspond to the linear transformation between their corresponding subspaces whose matrix representation is given by $\begin{bmatrix} -10 & -1 \\ 0 \end{bmatrix}$ with respect to the ordered bases $\{(1, 0, -1), (0, 1, 0)\}$ and $\{(1, 1, -1)\}$ for the domain and the codomain?

☐

$f(2+x, y, -x) = (2-x, 2+x, x)$

☐

$$f(2+x, y, -x) = (2+x, 2+x, x)f(2+x, y, -x) = (2+x, 2+x, x).$$

☐

$$f(2+x, y, -x) = (2-x, 2-x, x)f(2+x, y, -x) = (2-x, 2-x, x).$$

☐

$$f(2+x, y, -x) = (2-x, 2-x, -x)f(2+x, y, -x) = (2-x, 2-x, -x).$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(2+x, y, -x) = (2-x, 2-x, x)f(2+x, y, -x) = (2-x, 2-x, x).$$

Suppose the temperature $(T)(T)$ at a point $(x, y, z)(x, y, z)$ in $R^3 R^3$ is given by the function

$$T(x, y, z) = e^{x+y+z} T(x, y, z) = e^{x+y+z}$$

Let $T_1 T_1$, $T_2 T_2$, and $T_3 T_3$ be the tangent planes to the function $f(x, y) = \sqrt{9 - x^2 - y^2}$

$$f(x, y) = \sqrt{9 - x^2 - y^2}$$

at the points $(1, 2)(1, 2)$, $(2, 1)(2, 1)$, and $(2, 2)(2, 2)$, respectively. The set of points on each plane $T_1 T_1$, $T_2 T_2$, and $T_3 T_3$ form affine subspaces $A_1 A_1$, $A_2 A_2$, and $A_3 A_3$, respectively, of the vector space $R^3 R^3$, with respect to usual addition and scalar multiplication. Let $V_1 V_1$, $V_2 V_2$, and $V_3 V_3$ denote the vector subspaces corresponding to the affine subspaces $A_1 A_1$, $A_2 A_2$, and $A_3 A_3$, respectively. Answer the following questions using the above information.

3 points

Which of the following denotes the vector subspace $V_1 \cap V_2 V_1 \cap V_2$?

☐

$$\{(x, y, -x - y) \mid x, y \in R\} \{(x, y, -x - y) \mid x, y \in R\}$$

☐

$$\{(-2y - 2z, y, z) \mid y, z \in R\} \{(-2y - 2z, y, z) \mid y, z \in R\}$$

☐

$$\{(-2z, -2z, 3z) \mid z \in R\} \{(-2z, -2z, 3z) \mid z \in R\}$$

☐

$$\{(-2z, -z, z) \mid z \in R\} \{(-2z, -z, z) \mid z \in R\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(-2z, -2z, 3z) \mid z \in R\} \{(-2z, -2z, 3z) \mid z \in R\}$$

If $m m$ is the dimension of $V_1 V_1$, $n n$ is the dimension of $V_2 V_2$ and $r r$ is the dimension of $V_1 \cap V_2 V_1 \cap V_2$, then find the value of $m + n + r m + n + r$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

2 points

3 points

Which of the following denotes the maximum rate of change of temperature $(T)(T)$ at the point of intersection of the tangent planes $T_1 T_1$, $T_2 T_2$, and $T_3 T_3$?

☐

$$\sqrt{3}e^3 \quad 3$$

☐ $\sqrt{e^3}$

☐ $\sqrt[3]{e^{\frac{27}{5}}}$

☐ $\sqrt{e^{527}}$

☐ $\sqrt[3]{e^{\frac{3}{5}}}$

☐ $\sqrt{e^{53}}$

☐ $3e^{\frac{27}{5}}3e^{527}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

☐ $\sqrt[3]{e^{\frac{27}{5}}}$

☐ $\sqrt{e^{527}}$

2 points

Which of the following options is/are true?

☐

$T(x, y, z)$ is a linear function.

☐

$T(x, y, z)$ is continuous function over R^3 .

☐

$T_x(x, y, z)$ is not continuous function over R^3 .

☐

$T(cx, cy, cz) = (T(x, y, z))^c T(cx, cy, cz) = (T(x, y, z))c$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y, z)$ is continuous function over R^3 .

$T(cx, cy, cz) = (T(x, y, z))^c T(cx, cy, cz) = (T(x, y, z))c$

Suppose on approaching to the point $(1, 2, 3)$, temperature is approaching to e^a ea. What is the value of a ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

2 points

1 point

Let A be an $n \times n$ matrix such that $A^2 = 0$. Choose the correct statement from the following.

☐

Column space of A is a subset of the null space of A .

☐

Null space of A is a subset of the column space of A .

☐

Column space of A is equal to the null space of A .

☐

The given data is insufficient to conclude the relationship between the column space of AA and null space of AA .